

$$A = \begin{pmatrix} -1 & -2 & 0 \\ -1 & 3 & -1 \\ 2 & 3 & 1 \end{pmatrix}$$

char. polynom $p_A(\lambda) = \det(A - \lambda I)$

$$p_A(\lambda) = \lambda^3 - 3\lambda^2 + 4 = (\lambda - 2)^2(\lambda + 1)$$

$\lambda = 2$ - dim. vlastního prostoru = 1 $\Rightarrow$ jediné pod. bod 2
 $\lambda = -1$ - dimenze = 1

$$U_1 = \text{span} \{ \bar{v}, \bar{w} \}$$

$$\lambda = 2$$

$$(A - 2I)\bar{v} = 0$$

$$A - 2I = \begin{pmatrix} -3 & -2 & 0 \\ -1 & 1 & -1 \\ 2 & 3 & -1 \end{pmatrix}$$

$$\begin{aligned} -3x - 2y &= 0 \\ -x + y - z &= 0 \\ 2x + 3y - z &= 0 \end{aligned}$$

$$x = 2, y = -3, z = -5$$

podobný vektor $\bar{v} = (2, -3, -5)$

pročtený vektor $\bar{w}$ $(A - 2I)\bar{w} = \bar{v}$
 $(A - 2I)\bar{w} = (2, -3, -5)$

$$\begin{aligned} -3x - 2y &= 2 \\ -x + y - z &= -3 \\ 2x + 3y - z &= -5 \end{aligned}$$

$$x = 2, y = -4, z = -3$$

$$\bar{w} = (2, -4, -3)$$

$$\text{Basis } B_{U_1} = \{ \bar{v}, \bar{w} \}$$

$$A(\bar{v}) = 2\bar{v} + 0\bar{w}$$

$$(A - 2I)\bar{v} = \bar{v}$$

$$A(\bar{v}) = 2\bar{v} + \bar{v}$$

$$A(\bar{w}) = 1\bar{v} + 2\bar{w}$$

$$\text{pro } \bar{v} \quad A(\bar{v}) = 2\bar{v} + 0\bar{w}$$

$$\text{pro } \bar{w} \quad A(\bar{w}) = 1\bar{v} + 2\bar{w}$$

$$[A|_U]_{B_U} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

Smu rōj rāsting 'polymistor' $\lambda = 2$ a $\lambda = -1$

rāsting 'ektor' $\lambda = 2$: $\bar{r} = (2, -3, -5)$

rāsting 'ektor' $\lambda = -1$: $\bar{u} = (-1, 0, 1)$

$$U_2 = \text{Span} \{ \bar{r}, \bar{u} \}$$

Base $B_{U_2} = \{ \bar{r}, \bar{u} \}$

prod. matrix $\begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$